

UDIT KULKARNI

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Mumbai, India

PROFESSIONAL SUMMARY

Senior Software Engineer & Team Lead with a Master's in Computer Applications specializing in Machine Learning. Expert in building high-scale distributed systems and AI-native microservices using Python (LangGraph, FastAPI) and PHP (Laravel, Symfony, Drupal). Proven track record of leading technical teams for Tier-1 platforms like Disney+ Hotstar and architecting scalable RAG pipelines with Qdrant Vector Database. Deeply proficient in System Design, SOLID principles, and MVC patterns.

TECHNICAL SKILLS

- **Languages & Frameworks:** PHP 8.x (Laravel, Symfony, Drupal), Python (FastAPI, Flask), SQL, JavaScript.
- **AI & ML Engineering:** LangChain, RAG Pipelines, Qdrant Vector DB, Multi-Agent Orchestration, Prompt Engineering.
- **Infrastructure & Tools:** MySQL, Redis, Docker, AWS S3, Laravel Queues/Workers, Git, Composer, Drush.
- **Architecture:** System Design, OOP, Design Patterns (Repository, Factory), RESTful API Design.

WORK EXPERIENCE

► **Software Developer** Jan 2025 – Present
Publicis Digital Experience *HINCOL CRM Project*

- **AI-Driven CRM Intelligence:** Architected a Python microservice using LangGraph and Qdrant to enable intelligent data retrieval and multi-agent workflows. Built a Laravel integration layer to manage session state and broker AI requests.
- **Enterprise Account & Profile Management:** Developed core account headers, profiling models, and historical audit logs. Implemented Eloquent models and RBAC to manage complex customer hierarchies and metadata enrichment.
- **Sales Pipeline & Strategic Planning:** Engineered lead capture flows, opportunity management modules, and account planning repositories. Integrated lifecycle automation and event listeners for real-time status tracking.
- **Competitive Intelligence & Analytics:** Created competitor profiling and market positioning services that aggregate signals for trend analysis. Developed aggregation queries for CapEx visualization and positioning visuals.
- **Financial Performance & Reporting:** Designed calculation engines for time-series financial metrics and credit review metrics. Built advanced reporting flows and automated exports using scheduled background jobs.
- **CRM Operational Infrastructure:** Implemented issue resolution trackers with notification hooks, knowledge base search APIs, and geospatial persistence routines for region-based analytics.

► **Software Developer & Team Lead** Jan 2023 – Dec 2024
Razorfish *Disney+ Hotstar & Citroen*

- **Leadership (Disney+ Hotstar):** Directed squad progress for high-traffic streaming initiatives, ensuring delivery met schedule and quality standards. Built a custom project management tool to optimize internal developer velocity.
- **Pimcore & Symfony Engineering (Citroen):** Designed automated vehicle data-ingestion pipelines and integrated OpenID/OAuth authentication flows with third-party marketplaces including Magento 2 and Carwale.
- **Drupal & REST Ecosystem:** Developed custom resources for OTP verification, dealer locators, and Sales-force CRM synchronization. Managed S3 image path resolution for high-performance JSON:API delivery.
- **System Utilities:** Built PDF generation and reporting utilities using TCPDF. Assisted in Docker environment setup and documented composer-based installation procedures for distributed teams.

TECHNICAL FOCUS: AI & SYSTEM ARCHITECTURE

AI Microservice Architecture (Python / LangChain / Qdrant)

- Designed RAG pipelines featuring knowledge source vectorization and hybrid semantic/metadata filtering in Qdrant.
- Implemented SSE (Server-Sent Events) for streaming responses, reducing perceived latency in the bot interface.
- Developed complex conversation memory management, utilizing Redis for ephemeral context and persistent DB storage for long-term summaries.
- Integrated prompt injection protection and hallucination heuristics to ensure enterprise-grade safety.

Scalable Module Engineering (Laravel)

- **Design Patterns:** Utilized Service and Repository patterns to decouple business logic from API controllers, ensuring a testable and maintainable CRM codebase.
- **Asynchronous Processing:** Leveraged Laravel Queues for background tasks including batch embedding, report generation, and audit logging.
- **Performance Optimization:** Implemented database indexing, query caching, and DTO transformations for high-performance time-series data visualization.

EARLY CAREER

► Software Developer (Internship)

Jan 2022 – Jul 2022

Publicis Media

[View Certificate](#)

- Contributed to the backend development of an E-commerce platform using PHP.
- Focused on database schema design, modular code logic, and efficient data retrieval.

EDUCATION

► Master's in Computer Applications (Specialization: Machine Learning)

2021 – 2022

Sardar Patel Institute of Technology (SPIT), Mumbai

- Participated in Mumbai Avishkar Research Convention 2021; managed event logistics for the college.
- Volunteered for Research Colloquium and ICCICT-2021.

► Bachelor of Science in Computer Science

2017 – 2020

DG Ruparel College of Arts, Science and Commerce, Mumbai

PROFESSIONAL STANDARDS & ACHIEVEMENTS

- Expert in MVC, OOP, and SOLID principles for building maintainable, high-concurrency applications.
- Strong advocate for System Design best practices, including robust error handling and API versioning.
- Experienced in Dockerized development environments and automated testing with PHPUnit and PyTest.
- Proven ability to lead cross-functional teams and deliver complex technical projects for global clients.